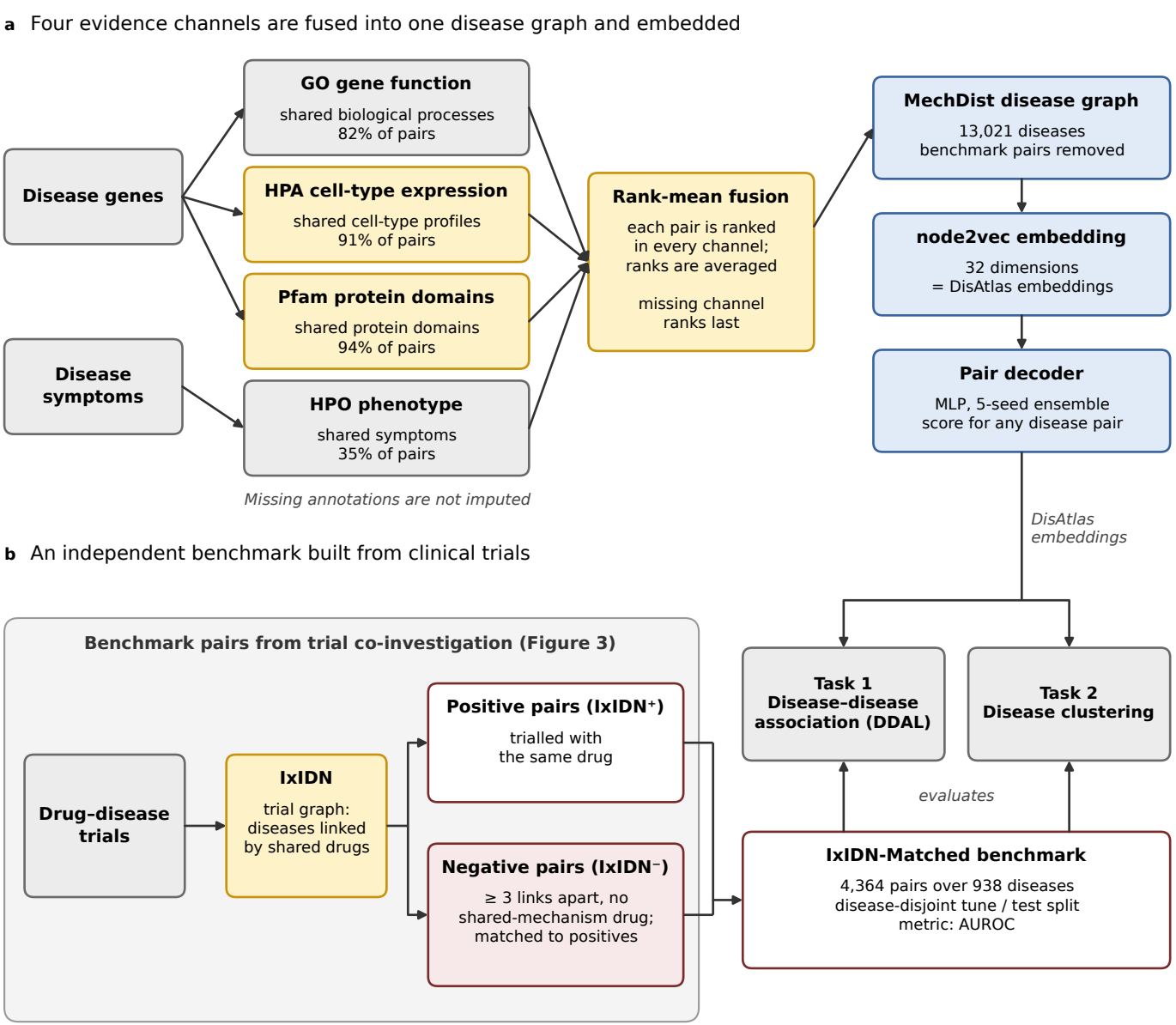
a Four evidence channels are fused into one disease graph and embedded
Disease genes
Disease symptoms
GO gene function
shared biological processes
82% of pairs
HPA cell-type expression
shared cell-type profiles
91% of pairs
Pfam protein domains
shared protein domains
94% of pairs
HPO phenotype
shared symptoms
35% of pairs
Missing annotations are not imputed
Rank-mean fusion
each pair is ranked in every channel; ranks are averaged
missing channel ranks last
MechDist disease graph
13,021 diseases
benchmark pairs removed
node2vec embedding
32 dimensions
= DisAtlas embeddings
Pair decoder
MLP, 5-seed ensemble
score for any disease pair
DisAtlas embeddings
b An independent benchmark built from clinical trials
Benchmark pairs from trial co-investigation (Figure 3)
Drug–disease trials
IxIDN
trial graph: diseases linked by shared drugs
Positive pairs (IxIDN+)
trialled with the same drug
Negative pairs (IxIDN−)
≥ 3 links apart, no shared-mechanism drug; matched to positives
Task 1
Disease–disease association (DDAL)
Task 2
Disease clustering
evaluates
IxIDN-Matched benchmark
4,364 pairs over 938 diseases
disease-disjoint tune / test split
metric: AUROC